

Model experiment of mold filling process in vertical centrifugal casting

Li Changyun^{*}, Wu Shiping, Guo Jingjie, Su Yanqing, Bi Weisheng, Fu Hengzhi

School of Materials Science and Engineering, Harbin Institute of Technology, P.O. Box 434, Harbin 150001, China

Received 29 September 2005; received in revised form 22 February 2006; accepted 12 April 2006

Abstract

By using centrifugal governor, high-speed camera, and liquid as well as acrylic glass molds to simulate the filling process of Titanium alloy melts. Results show that the liquid sticks to one side of the runner wall (or cavity wall) to fill and the side is opposite to the rotational direction. The whole filling process is divided into two sections, one is forward filling and the other is back filling. Experimental results show that bottom filling is better than top filling in that the former can realize stable filling, avoiding liquid crash in the cavity. The free surface of liquid is cambered like in the cavity and the runner, whose center is the rotational center.

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Keywords: Centrifugal casting; Filling ability; Hydraulic simulation; Bottom filling

1. Introduction

The 21st century is notable for the creation of a number of new structural materials among which titanium and its alloys are playing one of the leading roles [1]. Weight saving is the obvious one with titanium's high strength-to-weight ratio. The lower density of titanium compared with steel permits weight saving replacing steels even though they may be higher strength. Titanium could also replace aluminum when the operating temperature exceeds about 130°, which is the normal maximum operating temperature for conventional aluminum [2]. Corrosion resistance can be a very important issue. It resulted in rather wide application of titanium in chemical and shipbuilding industries. Now the use of titanium alloys in engineering structures for the extraction of hydrocarbons in offshore areas has become one of titanium applications [3]. With good mechanical properties and excellent biocompatibility, the use of titanium alloy has increased in dental application. In all, titanium alloy is used widely in many fields due to its good physical and chemical characters. The need of titanium alloys rises steeply these years, and the net-shape titanium alloy castings are now popular.

Majority of titanium alloy castings are complex thin-wall castings, often produced by vertical centrifugal casting, which uses centrifugal force to finish filling process and then obtain

castings to meet the needs [4–6]. Now more and more researchers and engineers study titanium alloys, but majority of the researchers focus on horizontal centrifugal casting and literature on vertical centrifugal casting is very poor [7–11]. Vertical centrifugal casting and horizontal centrifugal casting [12–14] in finishing stage of filling and solidification are conducted under high rotational speed of mold, so it is hard to study their filling process using conventional methods. Using the hydraulic simulation method, the whole filling process of vertical centrifugal casting was recorded by advanced high-speed camera. The filling regularities of vertical centrifugal casting were obtained by experimental results and theory analysis, which can offer theoretical basis for guiding production as well as improving production efficiency and quality.

2. Experimental procedure

The rotational velocity range of centrifugal machine used in the experiment is 0–1200 rpm, so different rotational velocities can be selected in the experiment and find which one is the best for filling. The high-speed camera used in the study can take 1000 (320 × 320) pictures per second, and can meet the need of the experiment. The experimental schematic diagram is shown in Fig. 1.

In order to study the filling process of liquid under centrifugal condition, two different filling methods were studied in the experiment: top filling and bottom filling. The schematic of the top gating system and bottom gating system are shown in Figs. 2 and 3, respectively. Three kinds of rotational velocities of mold (163, 245 and 375 rpm) were used in the experiment, and the rotational direction was anti-clockwise. Top filling and bottom filling were carried out at the same conditions. In order to assure the same initial velocity of liquid in the experiment, the liquid was poured out at the same height. The height was 50 mm, and the velocity of liquid reaching the bottom of sprue was 0.989 m/s.

^{*} Corresponding author. Tel.: +86 451 86418815.

E-mail addresses: lucy1226@126.com (L. Changyun), spwu@hit.edu.cn (W. Shiping), guojj@hit.edu.cn (G. Jingjie).

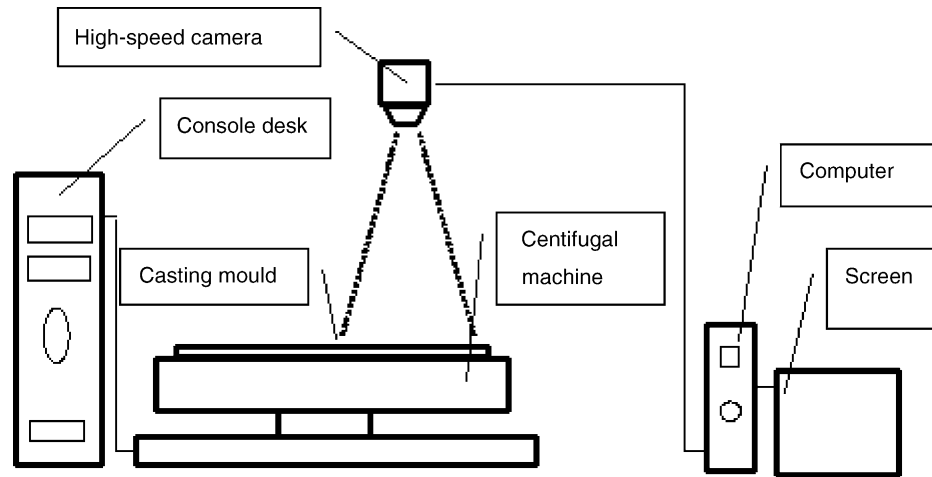


Fig. 1. Schematic diagram of hydraulic simulation in centrifugal casting.

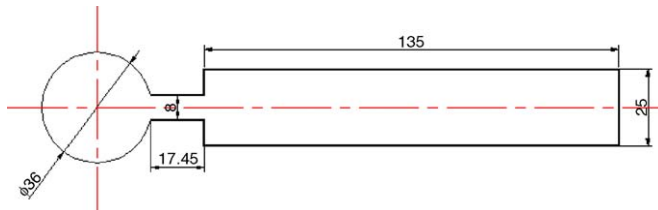


Fig. 2. Sketch of top filling and casing mould.

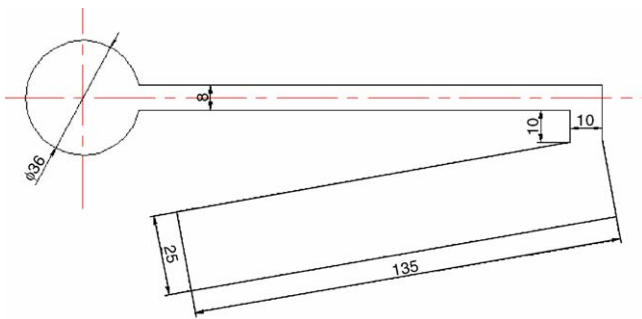


Fig. 3. Sketch of bottom filling and casing mould.

3. Experimental results and discussion

3.1. Phenomena appeared in the experiment

It can be seen from Fig. 4 that the filling process in the centrifugal condition is different from that in the gravity field. With

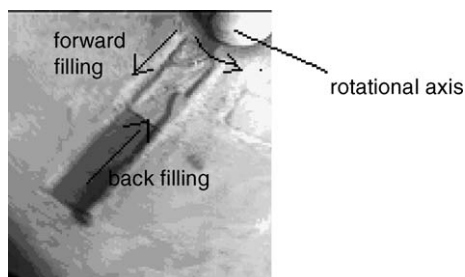


Fig. 4. Filling process of liquid in centrifugal filed.

the action of centrifugal force and Coriolis force, the liquid stick to the one side of the runner or cavity wall to fill. The wall is opposite to the rotational direction. The whole filling process is divided into two sections: forward filling and back filling.

Sheng [15] has discussed that the filling velocity increases with increasing filling length and rotational velocity. And the premise conditions of his discussion are: melt fills the cavity continuously; melt is in a fully developed steady state flow; dynamical viscosity coefficient μ and density ρ of melt are both constants; melt is regarded as an incompressible fluid; without regard to the friction loss during the filling process. Then he gave the equation of filling velocity at random position as follows:

$$v_m(t) = \frac{v_0 + \omega L_0}{4} e^{\omega t} + \frac{v_0 - \omega L_0}{4} e^{-\omega t} \quad (1)$$

where $v_m(t)$ is the filling velocity, v_0 the initial velocity, t the filling time, ω the angular velocity, and L_0 is the entrance length. What Sheng has discussed about the filling velocity was the limitation in the forward filling. It can be seen from photos taken in the experiment that the forward filling is an accelerated filling process, and the back filling is a uniform velocity filling process. Fig. 5 shows that the cross-sectional area is changeable during the forward filling, and is unchangeable during the back filling. It has been known that the flow of liquid is equal to the filling velocity and cross-sectional area product as follows:

$$Q = vs \quad (2)$$

where v is the filling velocity and s the cross-sectional area crossing by liquid. The flow Q is unchangeable at random time and the filling velocity increases with increasing filling length, so the cross-sectional area decreases with the increase of filling length. Fig. 6 gives the relationship between forward filling length and cross-sectional area during forward filling, and the calculation data shows good agreement with experimental data. During the back filling, the cross-sectional area is equal to the width of the cavity, and depends on Eq. (2), the filling velocity is unchangeable. Thus back filling is a uniform velocity filling process, which is different from the forward filling process.

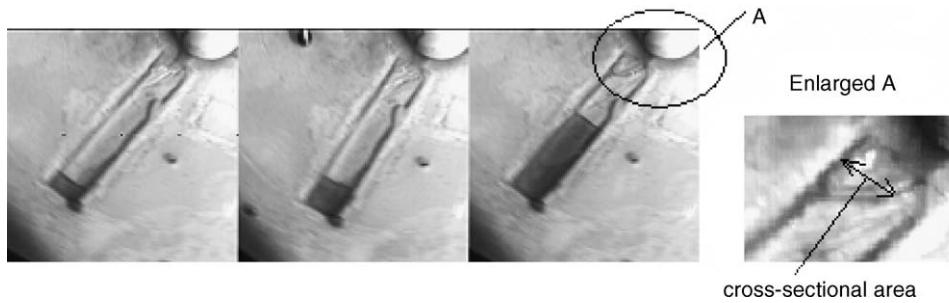


Fig. 5. Filling sequence of top filling at 245 rpm.

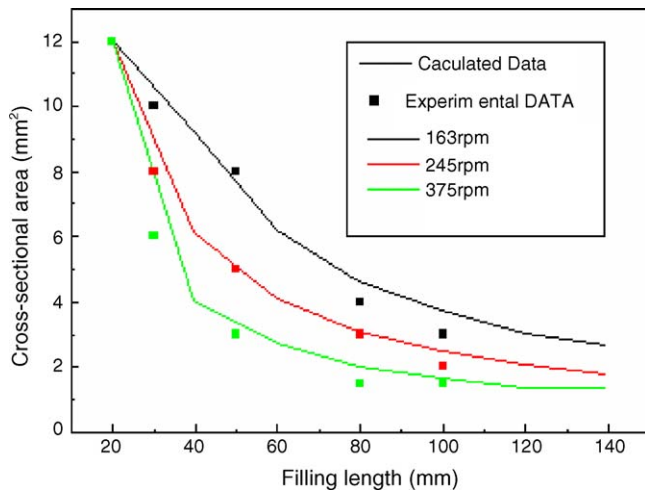


Fig. 6. Relationship between the forward filling length and cross-sectional area.

It can also be found from Fig. 7 that when the ingate is overflow, the whole filling system of bottom filling is divided into two sections, in the runner and in the cavity. And it can be seen from the photos that with the increase of filling time, the free surface of liquid in the runner and the cavity are not at the same level. Depending on the theory of equipressure surface, the free surface of liquid must at the same cambered surface under the action of the centrifugal force. Backing pressure is the main reason leading to this phenomenon, air exhaust in the runner is better than that in the cavity, and when liquid surface in the cavity heightens slowly, air pressure increases too, which will hinder the rising of liquid surface in reverse. This is the reason why the

surface level in the cavity is lower than that in the runner. And with the action of centrifugal force and Coriolis force the equipressure surface is cambered like whose center is the rotational center, that is why the free surface of liquid is cambered like in the cavity and the runner.

3.2. Comparisons between top filling and bottom filling

Figs. 5 and 7 show the filling process of top filling and bottom filling at 245 rpm, respectively. Because the two molds were put on the same rotational disk, the initial conditions were same for two filling methods. During the back filling, the free surface is cambered and the back filling length cannot be measured precisely. The back filling length in the cavity is equal to the average value of L1 and L2, as shown in Fig. 8. It is because of the same width and thickness of cavity, so only filling length can be used to consider the filling volume. Figs. 9 and 10 give the filling regularity of top filling and bottom filling at different rotational velocities (163, 245 and 375 rpm). Comparing Fig. 9 with Fig. 10, it can be found that the back filling length increases with increasing rotational velocity in both top filling and bottom filling. But the back filling velocity of liquid in the cavity is higher in top filling than that in bottom filling. There are two main factors leading to this phenomenon. One is the filling distance of liquid and the other is the backpressure. It can be seen from Fig. 3 that there is a runner (165 mm long) in bottom gating system, which increase the moving distance of liquid and it needs more time for liquid to reach the cavity, then it decreases the filling velocity of back filling in bottom filling. Another important

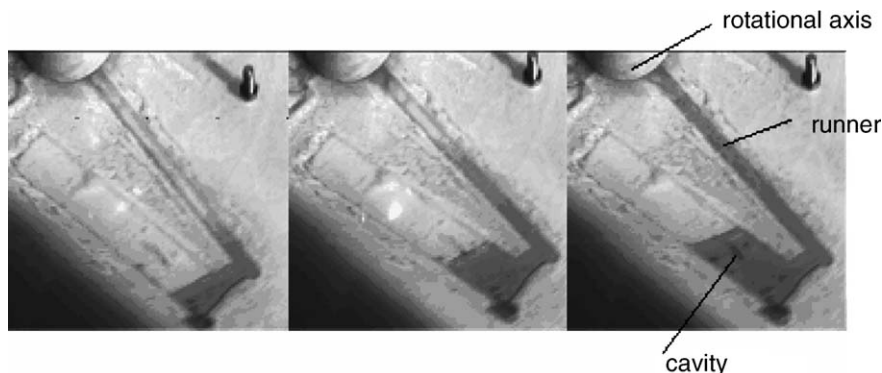


Fig. 7. Filling sequence of bottom filling at 245 rpm.

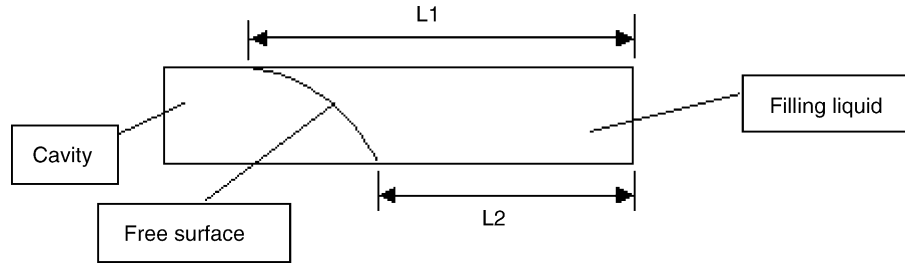


Fig. 8. Back filling length in the cavity.

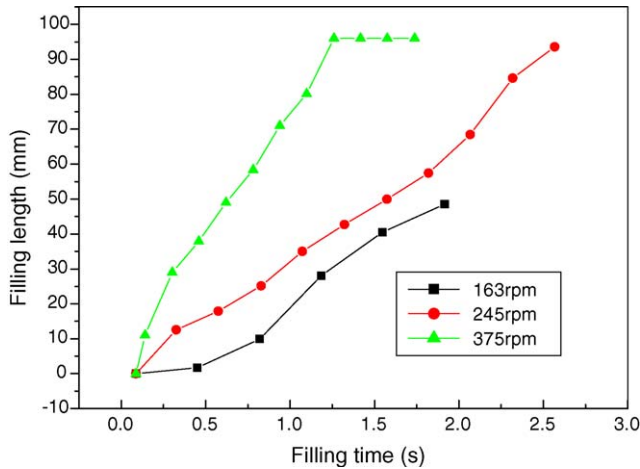


Fig. 9. Back filling length vs. filling time for top filling.

factor leading to the decrease of back filling velocity in bottom filling is backpressure. The experiment was not carried out in the vacuum condition, and when the liquid poured into the sprue, it entrapped gas into cavity. In the top filling, the entrapped gas was exhausted quickly from the runner inlet. However, in the bottom filling, when the liquid overflows the bottom of runner, the entrapped gas in the cavity is hard to emit from gas vent, which is smaller than runner inlet. Then, with increasing filling volume of liquid in the cavity, the backpressure increases too, which decreases the filling velocity of back filling inversely in bottom filling.

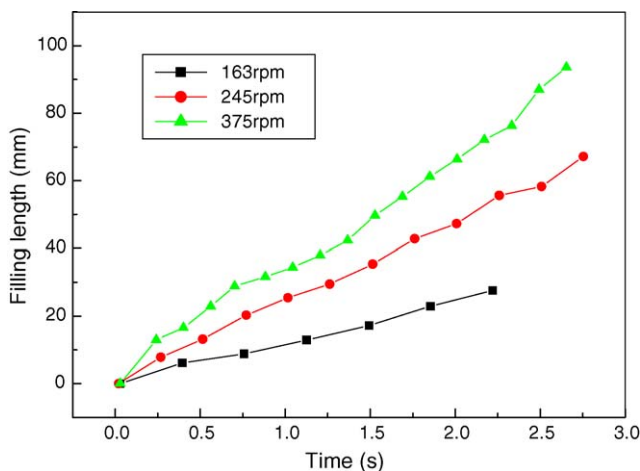


Fig. 10. Back filling length vs. filling time for bottom filling.

Although the back filling velocity in bottom filling is smaller than that in top filling, the bottom filling is better than top filling by comprehensive analysis. It can be found that the curves in Fig. 9 are more flat than that in Fig. 8, which indicates that the bottom filling is more stable than the top filling. Fig. 5 shows that the forward filling and the back filling were occurred in the cavity at the same time. Then the liquid crashes each other in top filling and the phenomena will continue until the cavity is full of liquid. The whole filling process is unstable in top gating system; the situation is different in bottom gating system. In the bottom filling process, the liquid crash phenomena occurred in the runner, and when the runner bottom is overflowed cavity filling process becomes stable. It has been known, that the casting defects often exist in the final solidified zone. With the action of centrifugal force and Coriolis force, the final solidified zone occurred at the runner inlet in vertical centrifugal casting. Then the defects occur in the runner in the bottom filling process, but the runner in the top gating system is short and the final solidified zone will exist in the cavity. Taking into account that the casting of titanium alloy are often carried out in vacuum condition, back pressure has a little effect on filling velocity and the filling velocity can be increased by increasing the rotational velocity. The bottom filling method is better than top filling method because the former can realize stable filling in the cavity and can reduce the defects forming in the cavity, which is important for the production of titanium alloy casting.

4. Conclusions

- (1) Hydraulic simulation experiments were studied in this paper, including two kinds of filling methods (top filling, bottom filling) and three kinds of rotational velocities (163, 245 and 375 rpm). Photos taken in the experiment show that in both top filling and bottom filling, liquids stick to back-wall of cavity or runner to filling due to the action of centrifugal force and Coriolis force. Filling volume is rising with the increase of the filling time and rotational velocity of mold.
- (2) The whole filling process is divided into forward filling and back filling. During the forward filling, liquid cross-sectional area decreases with the increase of forward filling length and rotational velocity. But cross-sectional area is unchangeable in the cavity during the back filling. Forward filling is an accelerated velocity process and back filling is a uniform velocity filling process. In the whole filling process

of the experiment, backpressure has a great effect on filling velocity and then it needs to increase centrifugal force to obtain complete castings.

- (3) The experiments show that the bottom filling method is better than the top one, which can achieve stable filling, minimize turbulence and avoid drastic liquid collision.

Acknowledgement

Authors would like to thank Project 973 (No. 51319) for financial support.

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